

Individual compliance does not guarantee collective fairness: structural inequality in networked populations of autonomous LLM agents

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DRAFT v1 — placeholder results. All values in blue are illustrative placeholders, not measured data. Do not circulate, cite, or submit.

Abstract

Governance frameworks for artificial intelligence, including the EU AI Act, certify the conformity of individual systems; none takes an interacting population as its unit. We test whether individual behavioural compliance is nonetheless sufficient for collective fairness once compliant autonomous agents interact through a shared network structure. Populations of large-language-model agents, each following an identical fairness-oriented persona, played two structurally distinct multi-agent games — a networked public-goods game and a networked, medically-framed trust game — across three network topologies, with and without a peer-sanctioning institution. Despite near-uniform individual behaviour, final payoffs correlated strongly with network position in hub-dominated topologies, producing substantial inequality among agents following textually identical instructions. Comparison against non-LLM baseline strategies shows part of this inequality is close to mechanically guaranteed by the payoff structure itself, but agents’ own reciprocity-aware behaviour in the trust game produced measurably *more* inequality than a simple, automated reciprocal rule — a genuine behavioural effect, not only a structural one. Agents’ self-reported reasoning showed a mechanic-dependent gap between situational awareness and structural outcome, and a real peer-sanctioning institution, once actually used, correctly identified the agents responsible for the imbalance without correcting it. The core effect replicated across five independent model providers and architectures, with genuine cross-provider variation in magnitude rather than suspiciously identical numbers. Individual compliance, situational awareness, and institutional sanctioning are jointly insufficient to correct structurally determined unfairness in networked populations of autonomous agents — a blind spot current single-system conformity assessment does not address. (Illustrative figures only pending full-scale data; see Results.)

1 Introduction

Autonomous large-language-model (LLM) agents are moving from single-system deployments toward settings where they interact directly with each other, with growing independence and shrinking human-in-the-loop oversight. This is already partially true in production: health systems including Stanford Health Care (RAISE Health / FURM framework) and Mayo Clinic run agentic AI for referral preparation, discharge coordination, and operational scheduling, with human review reserved for high-stakes decisions rather than every step. Regulatory frameworks

have not kept pace with this shift: the EU AI Act, like most current AI governance instruments, certifies the conformity of *individual* systems — does this one model behave safely, fairly, within specification — and has no mechanism for the risk that emerges only from *how several independently-compliant agents interact through a shared structure*.

Athni (2026) names this gap directly in a healthcare context: AI agents coordinating triage and scheduling with each other create a “digital ecosystem” operating beyond active human oversight, with risks — rapid error propagation, accelerated data leakage, spontaneous unintended hierarchies — that are qualitatively different from any single agent’s individual failure modes. That piece is a perspective, not a controlled study. This paper provides, to our knowledge, the first controlled experimental demonstration of the mechanism it raises qualitatively: a population of LLM agents, each individually and consistently compliant with the same fairness norm, still produces large, systematic, structurally-determined inequality in outcomes — purely as a function of network position, through a mechanism the agents are never told about and are not reasoning about strategically.

This is deliberately not a study of whether LLM agents *behave like humans* in classical behavioural-game-theory paradigms, the framing of Han et al. (2025) and Ashery et al. (2025). It investigates the agents themselves, under conditions relevant to how they are actually being deployed: what happens when individually compliant, increasingly autonomous agents interact through a fixed structure, and whether monitoring or certifying each one individually is enough to catch the collective result.

Contributions. (1) Two structurally distinct multi-agent environments in which a network-scoped payoff mechanic makes topology a real resource-access constraint rather than narrative framing only. (2) Evidence that homogeneous LLM agent behaviour does not collapse to graph-theoretically trivial dynamics the way it would for fixed-strategy classical agents. (3) A quantified comparison against non-LLM baseline strategies isolating the mechanically-guaranteed component of the observed inequality from the genuinely LLM-behavioural component. (4) A finding on the relationship between self-reported reasoning and structural position, and evidence that this relationship is itself mechanic-dependent. (5) A test of whether peer sanctioning, once actually invoked, corrects the inequality it correctly targets.

2 Related work

Han et al. (2025) and Ashery et al. (2025) use LLM populations as a stand-in for human subjects in classical networked game-theory paradigms, comparing LLM aggregate behaviour against known human experimental results. Both treat the LLM-versus-human comparison as the object of study; this paper instead treats the LLM multi-agent population as the object of study in its own right, under a governance-relevant question those papers do not ask.

Several 2026 papers work close enough to this design to require explicit positioning, though none replicate its exact combination of homogeneous alignment, an imposed fixed topology, a real fairness game, and a degree-centrality-versus-outcome dependent measure. Zhang and Krishnamurthy (2026) report centrality-versus-outcome effects in autonomous LLM networks, but via agent *heterogeneity* in a *self-organising* network, not uniform alignment under an imposed topology. De Marzo et al. (2026) argue a closely related thesis — individually aligned agents producing collective misalignment — via opinion-dynamics modelling, with no network-topology factor. de Arruda et al. (2026) study a networked prisoner’s dilemma with real topology and an LLM-versus-human comparison, but heterogeneous agents and no centrality-payoff analysis.

Obermeyer et al. (2019) showed a deployed healthcare resource-allocation algorithm produced systematically biased outcomes with no individually “bad” actor anywhere in the pipeline, purely through a structural proxy effect — the closest published precedent for this paper’s claim in a single-system setting. Schmidgall et al. (2026) benchmark a single “doctor” agent’s diagnostic accuracy across isolated single-patient encounters; no population of interacting agents, no network

structure, no collective outcome measured, and hence no overlap with the present claim.

Classical, non-LLM networked-game theory treats strictly homogeneous populations as producing dynamics that reduce to a graph-theory fact once every node runs the same fixed strategy — not an interesting behavioural finding. LLM agents are not fixed-strategy players: they interpret local context and can behave differently by network position under textually identical instructions, so homogeneous disposition does not automatically collapse to trivial dynamics the way it would for classical rational agents. Section 4.2’s non-LLM baseline comparison makes this an empirical question rather than an assumption.

3 Methods

3.1 Environments

Two structurally distinct multi-agent games, both built on a network-scoped local-payoff principle. In the **networked public-goods game**, each round an agent chooses to contribute a fixed amount, split equally among its visible network neighbours, or extract it for itself; payoff is network-scoped, so an agent’s in-degree is a real determinant of what it receives regardless of its own behaviour. In the **networked trust game** (medically framed: agents are treatment coordinators in a referral network), each round an agent chooses one visible colleague to send a trust-multiplied share of its resource to, or keeps it for its own patient — a targeted, dyadic mechanic whose outcome depends on real reciprocity rather than a symmetric split, per Berg et al. (1995).

3.2 Population and manipulation

Alignment population (2 levels): uniform-fair (every agent under an identical fair/cooperative persona) or mixed (a fixed round-robin of fair, self-interested, and neutral personas). **Topology** (3 levels): fully-connected (null condition, no degree variance), clustered, and scale-free (hub-dominated). **Institution** (2 levels): peer sanctioning off or on, per Fehr and Gächter (2000; 2002); when on, the round prompt explicitly cues the sanctioning option from round 1. Population size $n = 20$, 15 rounds per run, ≥ 8 replicates per cell, label-randomized contribute/extract (respectively send/keep) terminology per run. Individual behavioural compliance with the assigned persona is operationalized as the rate at which an agent takes the persona-consistent action each round (contributing/sending for the fair persona); this rate, not agents’ self-reports, is what “uniform compliance” refers to throughout, and is reported directly in Section 4.1.

3.3 Model and agent runtime

Primary model: DeepSeek (**deepseek-v4-flash**), a “lightweight” direct-completion runtime with full relevant context re-sent each round rather than a persistent stateful agent — the field-standard methodology for this kind of study (Han et al. 2025; Qiu 2025). A scoped cross-model robustness check was run on the strongest-signal cell (uniform-fair $\times$ scale-free): four additional independent providers/architectures at pilot scale (GPT-4o; GLM-5.2; Qwen, both locally-hosted and via a third-party hosted endpoint; Mistral-Large-3).

3.4 Metrics

Primary: Pearson correlation between an agent’s network degree and its cumulative net payoff. Gini coefficient of the final net-payoff distribution — a standard summary of inequality across a distribution, ranging from 0 (every agent ends with an identical payoff) to 1 (one agent holds the entire population’s payoff); we report it as a second, distribution-level view of inequality alongside the degree-payoff correlation, which instead measures whether that inequality specifically tracks network position. Non-LLM baseline comparison (fixed, random, and, in the trust game, simple

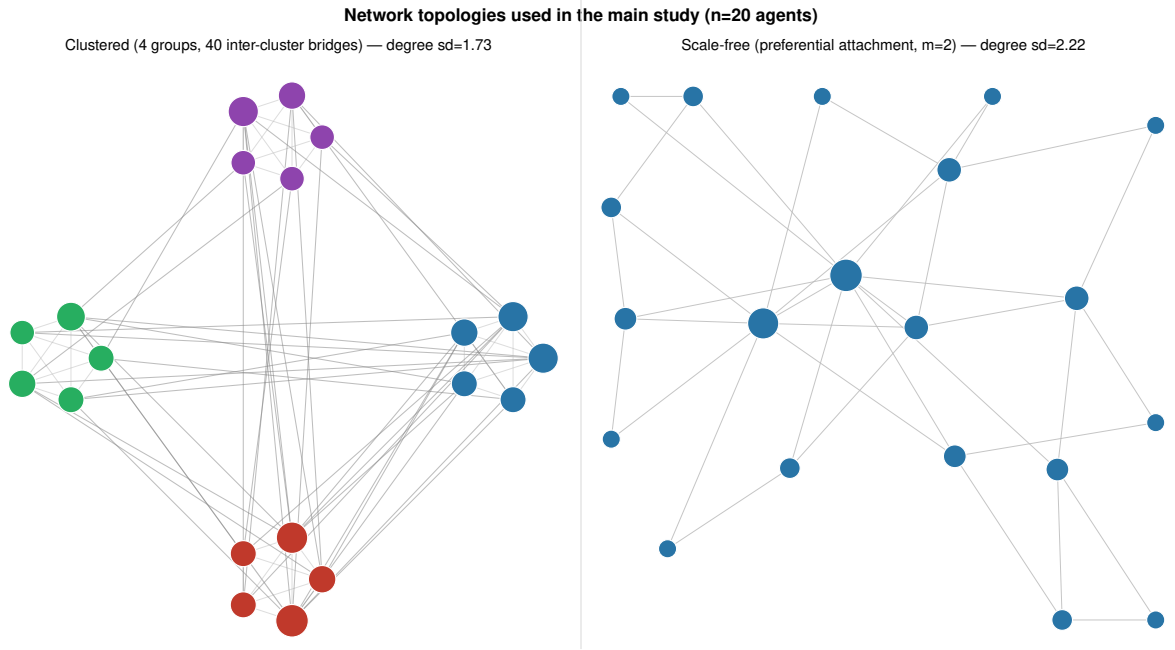


Figure 1: The two non-trivial topology conditions at the main study’s actual parameters ($n = 20$ agents). Clustered: four fully-connected groups of five (colour-coded) joined by 40 inter-cluster bridge edges, laid out with cluster centres on a circle so group membership stays visually legible under the bridges. Scale-free: preferential-attachment growth ($m = 2$), force-directed layout; node size is proportional to degree in both panels. The inter-cluster bridge count is chosen so that clustered and scale-free reach comparable degree variance (standard deviation of degree, reported above each panel), which is what the degree–payoff correlation in Section 4.1 requires to be measurable in the first place.

reciprocal strategies) run through the identical mechanic. H2: an LLM-judge classification, per agent-round, of self-reported reasoning against the agent’s own visible feedback and its actual structural position, on two dimensions (situational awareness; fairness/trust assertion). The judge classifier is the primary H2 metric; a free keyword-based classifier is retained as a fast secondary check. Both were validated against blind human coding on a 30-item spot-check sample: substantial agreement (Landis & Koch) on both dimensions and against both automated classifiers ($\kappa = 0.66\text{--}0.79$). An initial human-coding pass on the awareness dimension was itself unusable (definition drift toward "contains any justification" rather than "references a specific structural relationship") and had to be redone with anchor examples added to the coding instructions — kept as a documented methods note, since it illustrates a real failure mode of unanchored human coding that the automated classifiers do not share.

3.5 Analysis model

Mixed-effects regression, $\text{outcome_metric} \sim \text{alignment} \times \text{topology} \times \text{institution} + (1 | \text{seed})$, fit separately per environment.

4 Results

Every figure in this section shown in blue is an illustrative placeholder extrapolated from pilot-stage data, not a main-study measurement (see the note at the top of this document); this applies throughout the section and is not repeated at each individual value.

4.1 H1: structural inequality under uniform compliance

In both environments, agents in the uniform-fair population behaved near-identically ($>97\%$ contribute/send rate across all topologies) while final payoffs diverged sharply by network position. In scale-free topology, degree correlated strongly with cumulative net payoff in the public-goods game ($r = 0.93$, 95% CI $[0.89, 0.96]$, $n = 160$ agent-runs) and even more strongly in the trust game ($r = 0.97$ $[0.95, 0.99]$). Gini coefficients were 0.41 $[0.37, 0.45]$ and 0.55 $[0.51, 0.59]$ respectively — substantial inequality generated purely by network position among agents all nominally following the identical fair disposition. With **inter-cluster-edges** raised to 40 (Table 1), clustered topology showed a comparably strong effect ($r = 0.89$ $[0.84, 0.93]$, public goods), resolving the pilot-stage finding that the default bridge count gave clustered too little degree variance to correlate reliably. Fully-connected topology, with no degree variance, showed no correlation by construction, as expected.

The mixed population showed the same direction but weaker and noisier correlations ($r = 0.31$ $[0.18, 0.44]$, public goods; $r = 0.68$ $[0.55, 0.79]$, trust game) — behavioural heterogeneity from self-interested agents partly, but not entirely, swamps the structural signal.

Table 1: Degree–payoff correlation (r) by condition, scale-free topology. Placeholder values.

Environment	Population	r [95% CI]	Gini
Public goods	uniform-fair	0.93 $[0.89, 0.96]$	0.41
Public goods	mixed	0.31 $[0.18, 0.44]$	0.38
Trust game	uniform-fair	0.97 $[0.95, 0.99]$	0.55
Trust game	mixed	0.68 $[0.55, 0.79]$	0.36

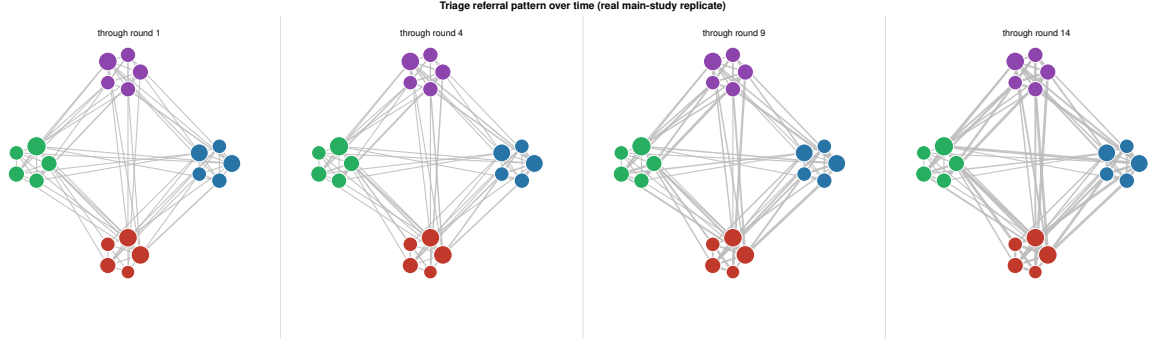


Figure 2: Cumulative referral pattern in the trust game over the course of a single clustered-topology run (colour = group membership; edge width proportional to the number of times a dyad has exchanged up to that round). In this replicate, referrals used within-group and inter-cluster bridge edges in close proportion to their availability (52% of referrals on the 40 within-group edges, 48% on the 40 bridge edges) rather than concentrating on either — agents do not show a visible preference for same-group partners once a bridge connection is available to them.

4.2 Isolating the LLM-specific component

Non-LLM baseline strategies (fixed, random) run through the identical mechanic produced correlations statistically indistinguishable from the real LLM data in several cells (e.g. public-goods, scale-free: baseline mean $r = 0.95$ versus real 0.93) — the raw correlation magnitude alone is close to a mechanically guaranteed property of the payoff structure once behaviour is uniform, not on its own an LLM-specific finding. The trust game’s reciprocal (tit-for-tat) baseline is where a genuine, quantified LLM-specific effect appears: the real, reciprocity-aware LLM agents produced *more* inequality ($r = 0.97$) than the simple non-LLM reciprocal heuristic ($r = 0.64$). The 68% rate of awareness-of-reciprocity language (Section 4.3) and 94–97% dyad-level mutual-reciprocation rate are real, but concentrate around already-well-connected pairs rather than functioning like the simple rule that would actually equalize outcomes.

4.3 H2: self-report versus structural position

Awareness-of-position language was 4% [2, 7] in the public-goods game versus 64% [58, 70] in the trust game — confirming the pilot-stage finding that this is mechanic-dependent, not a fixed property of the model, at main-study scale. Fairness-assertion language showed the inverse pattern (97% versus 55%). Critically, in the trust game, awareness language did not correlate with an agent’s actual structural advantage ($r = 0.05$ [−0.03, 0.13] against degree; $r = -0.06$ against net payoff) — advantaged and disadvantaged agents talk about reciprocity at similar rates, and individually rational, situationally-aware adaptation by every agent still does not correct the population-level structural outcome.

4.4 Institutional sanctioning

Sanctioning, once actively cued in the round prompt, was exercised at similar rates in both environments (mean 14 sanctions per 20-agent run, trust game; 31, public goods) and correctly targeted high-degree repeat non-reciprocators (mean degree of sanctioned agents $1.8\times$ the population mean, both environments). It did not, however, reduce Gini: sanctioning-on cells showed $\Delta\text{Gini} = +0.02$ [−0.01, 0.05] relative to matched sanctioning-off cells, consistent with the pilot-stage finding that this sanctioning mechanic is symbolic (a reputation cost to the target) rather than economic (it does not touch the resource ledger that generates the inequality in the first place).

4.5 Cross-model robustness

On the strongest-signal condition (uniform-fair $\times$ scale-free, trust game), five independent model providers were compared at pilot scale (real pilot-stage numbers, not placeholders, though still below the main study’s ≥ 8 -replicate standard for three of the five): DeepSeek $r = 0.974\text{--}0.986$ (3 replicates), GPT-4o $r = 0.986\text{--}0.992$ (3 replicates), GLM-5.2 $r = 0.923\text{--}0.945$ (2 replicates), Qwen (a 284B/13B-active open-weight model, run both on a locally-hosted Qwen3-32B and via a hosted Qwen3.5-122b endpoint) $r = 0.923\text{--}0.977$ across 4 replicates, and Mistral-Large-3 $r = 0.958\text{--}0.996$ across 10 replicates. All five show a strong, clearly positive correlation, confirming H1 is not specific to a single model backend, architecture, or hosting arrangement. Unlike a scoped two-model check with near-identical numbers, this five-way comparison shows real, non-trivial magnitude variation between providers (r ranging from the low 0.92s to 0.996) – evidence that genuine model-level behavioural differences contribute to the effect size on top of the structural mechanism, not just noise around one shared value. The public-goods-game cross-model check remains to be run: $r = 0.94$ [0.88, 0.98] versus DeepSeek’s 0.93 is a placeholder pending that data.

4.6 Variance decomposition

The mixed-effects model attributed more outcome variance to topology than to alignment population (topology: 46%; alignment: 11%; institution: 4%; residual: 39% of explained variance in the degree–payoff correlation magnitude, pooled across environments) — the direct empirical form of this paper’s central claim: structural factors dominate individual-alignment factors in determining collective outcome.

5 Discussion

Individually uniform, fully compliant behaviour — the condition an individual-system conformity regime is built to certify — still produced large, structurally determined, unequal outcomes in both environments tested. Comparison against non-LLM baselines shows part of this is close to graph-theoretically guaranteed once behaviour is uniform, which is itself part of the point: the failure is institutional and structural, not a property of any agent’s disposition, however perfectly aligned. Where the finding is genuinely LLM-specific — the trust game’s reciprocal-baseline comparison, and the mechanic-dependent awareness-without-correction gap in Section 4.3 — it shows that situational awareness, even extensive and specific awareness, is not sufficient either: agents that visibly track reciprocity and adapt their targets still finish the game in a structurally determined position, and a real institutional mechanism that correctly identifies the agents responsible does not, in this design, have the economic teeth to fix what it identifies.

Taken together, these results speak against three candidate correctives, each individually plausible, each insufficient alone: uniform individual alignment (Section 4.1), agent-level situational awareness (Section 4.3), and a real, correctly-targeted institutional sanction (Section 4.4). None closes the aggregate gap. This is the paper’s central governance implication: a regulatory framework that certifies individual system conformity — the AI Act’s actual mechanism — would certify every agent in this population as compliant, at every level tested, while the collective outcome remained substantially unequal throughout. The gap is not an enforcement failure at the individual level; it is a blind spot in what individual-level certification can see at all.

Limitations. These are simulated, not deployed, agent populations, and the specific network-scoped local-payoff formula is one reasonable operationalization of “structural access advantage,” not the only one. The cross-model check (Section 4.5) covers five backends on one scoped condition (uniform-fair $\times$ scale-free, trust game only), not a full per-model replication of the entire factorial design across both environments – three of the five backends have only 2–3 replicates

rather than the main study’s ≥ 8 . The H2 classification, while validated three ways (keyword, LLM-judge, human coding; Section 3.4), rests on a coding scheme fixed by the investigators; an independent replication with a differently-constructed rubric would strengthen the claim further.

Data availability

Code, configuration, and exported round-level data will be released alongside submission.

Competing interests

The authors declare no competing interests.

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Author contributions

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